Introduction to Robotics

Lecture 2: Transformations

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Template made by Panagiotis Papagiannopoulos

Logistics

Laboratory of Automation & Robotics

- **Lectures**:
  - 12 Lectures
  - 1 Recitation Lecture
  - Lab
- **Examination**:
  - 4 Homeworks **(30% of total grade)**
  - Oral Exam **(50% of total grade)**
  - Lab **(20% of total grade)**
- **Office Hours**:
  - **Tue-Thu** *(10:00-12:00)*
  - 24/7 by email (`costashatz@upatras.gr`, subject: *ECE_IR_AM*)
- **Material and Announcements**

What is this course about?

All the basics for delving into the field of Robotics:

- Rigid Body Motions in 2D and 3D
- Homogeneous Transformations and Exponential Coordinates
- Forward Kinematics
- Inverse Kinematics
- Jacobians and Singularities
- Velocity Control
- Trajectory Generation
- Dynamics of a Single Rigid Body and Open-Chains
- Force/Torque Control

Lectures/Lab Format

- **3-hour lectures:**
  - Theory
  - Hints for personal reading!
  - Live code examples!
  - Ask questions please!
- **2-hour labs (in teams):**
  - **20% of total grade**!
  - ROS
  - Arduino
  - Work on real systems!

Transformations

- A rigid body in 3D space needs at least six values to completely determine its pose[1]
- But where is the body's pose? With respect to what?

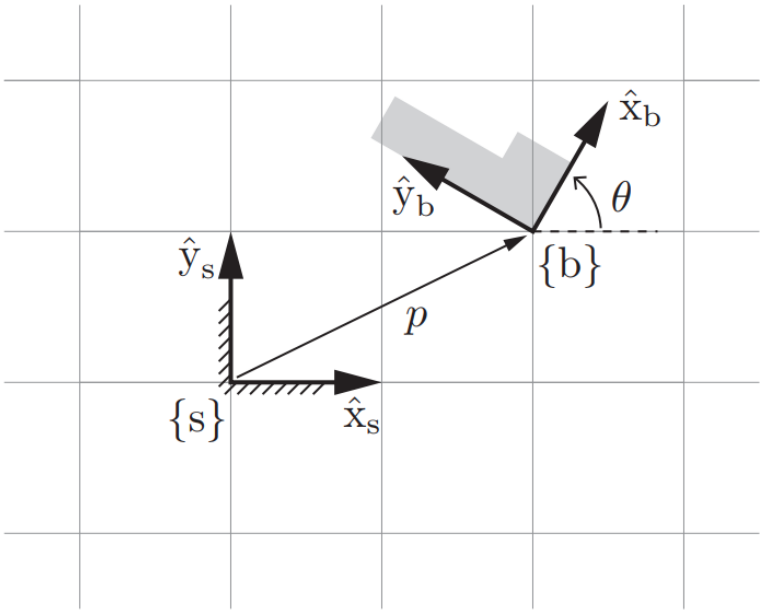
[1] We need at least 3 for bodies in 2D space.

Transformations

- A rigid body in 3D space needs at least six values to completely determine its pose[1]
- But where is the body's pose? With respect to what? We need a *reference frame*
- We always assume a static reference frame that does not change or move at all: we usually refer to it as *"world frame"*
- Each body has its own local static reference frame (It can have more than one!): we usually refer to it as *"body frame"*

[1] We need at least 3 for bodies in 2D space.

Transformations in 2D Space (1)



- Let's assume that $\{s\}$ is the world frame, and that $\{b\}$ is the body frame. How can we represent the body frame $\{b\}$ with respect to $\{s\}$?

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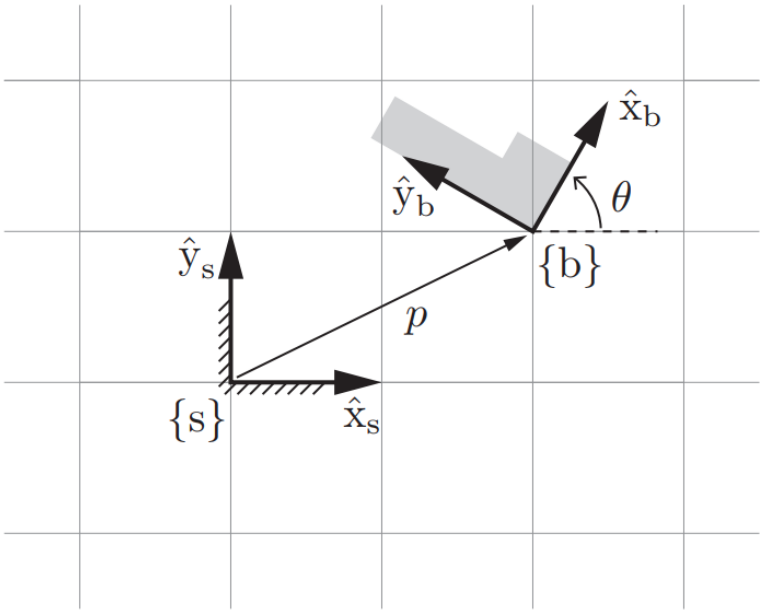

- Let's assume that $\{s\}$ is the world frame, and that $\{b\}$ is the body frame. How can we represent the body frame $\{b\}$ with respect to $\{s\}$?
- $\boldsymbol{p} = \boldsymbol{p}_x^s \hat{\boldsymbol{x}}_s + \boldsymbol{p}_y^s \hat{\boldsymbol{y}}_s$

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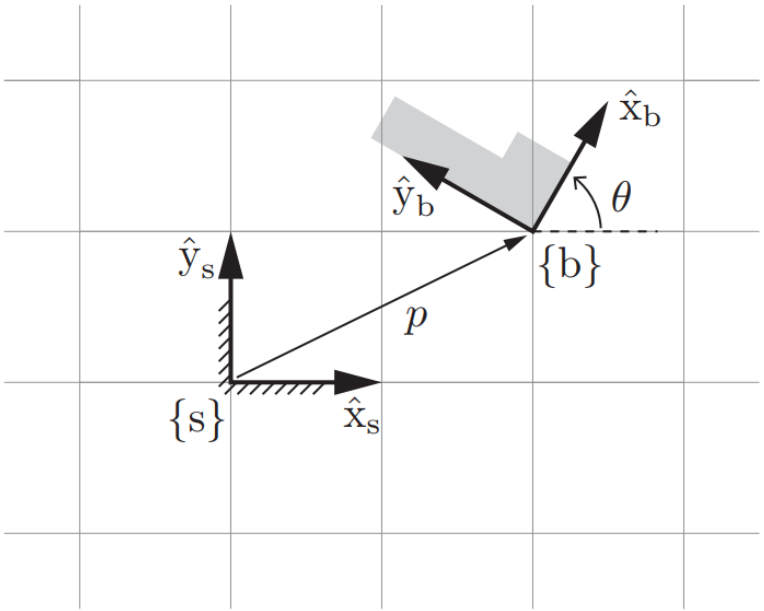
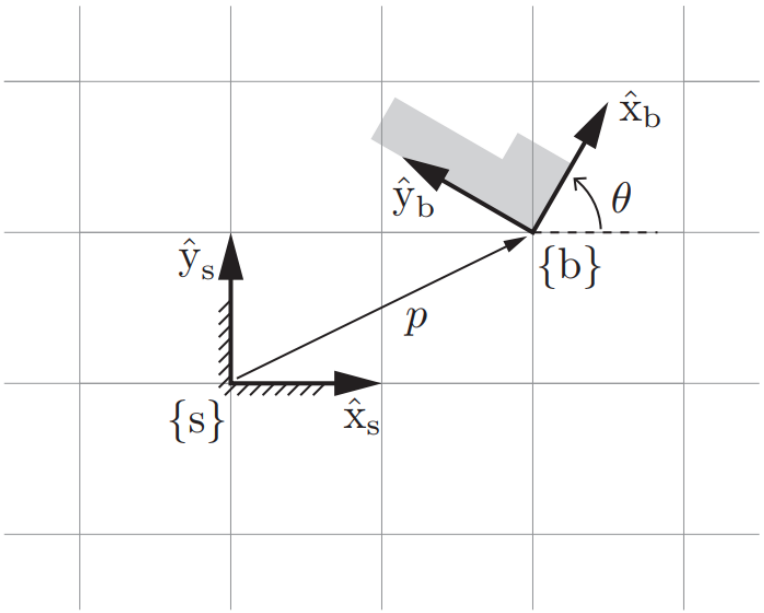


- Let's assume that $\{s\}$ is the world frame, and that $\{b\}$ is the body frame. How can we represent the body frame $\{b\}$ with respect to $\{s\}$?
- $\boldsymbol{p} = \boldsymbol{p}_x^s \hat{\boldsymbol{x}}_s + \boldsymbol{p}_y^s \hat{\boldsymbol{y}}_s$
- $\hat{\boldsymbol{x}}_b = cos\theta \hat{\boldsymbol{x}}_s + sin\theta \hat{\boldsymbol{y}}_s$
- $\hat{\boldsymbol{y}}_b = -sin\theta \hat{\boldsymbol{x}}_s + cos\theta \hat{\boldsymbol{y}}_s$

Source: Modern Robotics: Mechanics, Planning, and Control, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press.

Transformations in 2D Space (2)



With Vector/Matrix Operations:

- $\boldsymbol{p} = \begin{bmatrix} \boldsymbol{p}_x^s \\ \boldsymbol{p}_y^s \end{bmatrix} \in \mathbb{R}^2$

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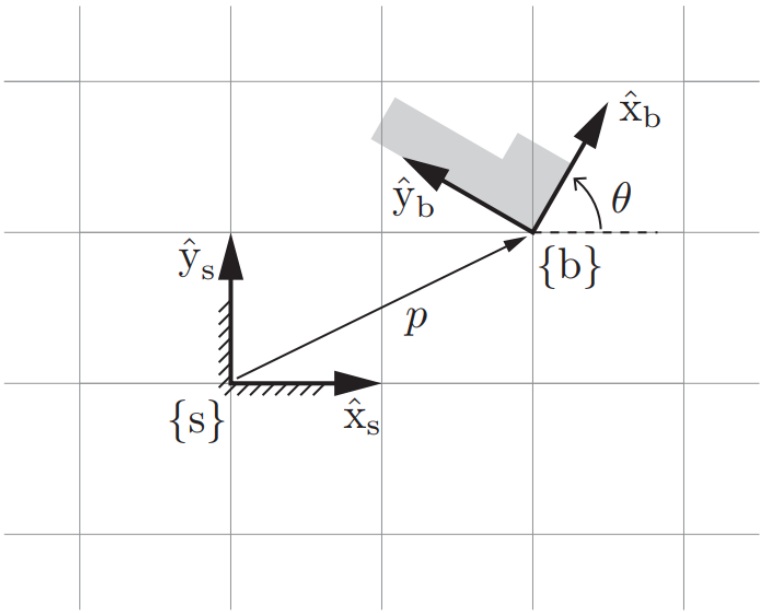

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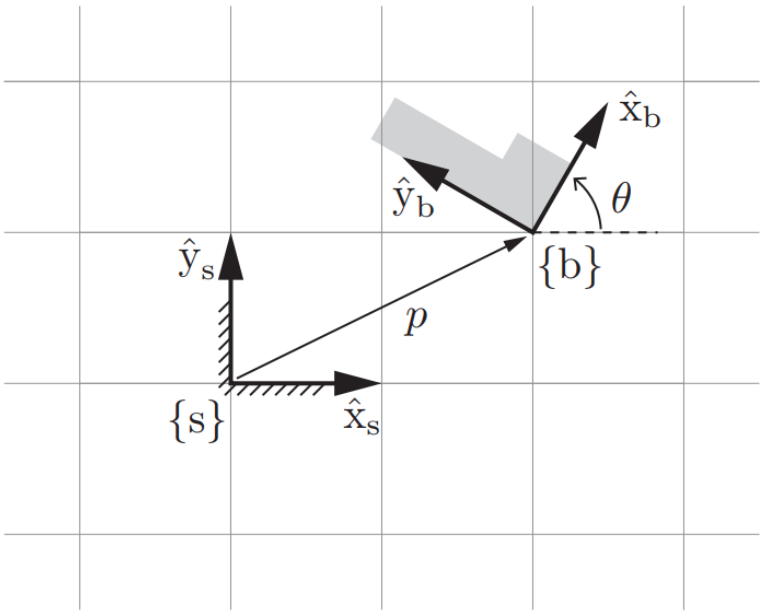

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- $\boldsymbol{R}$ is called the rotation matrix
- The tuple $(\boldsymbol{p}, \boldsymbol{R})$ represents the transformation from the body frame to the world frame

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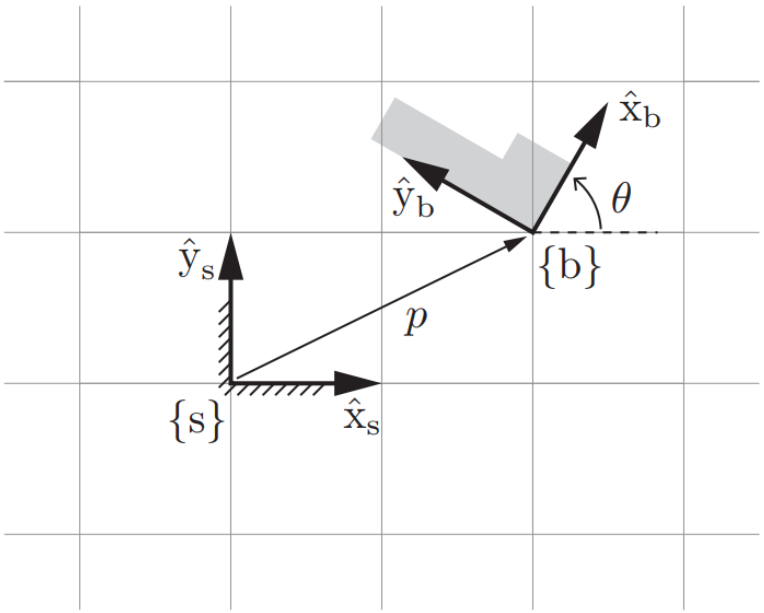

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- $\boldsymbol{R}$ is called the rotation matrix
- The tuple $(\boldsymbol{p}, \boldsymbol{R})$ represents the transformation from the body frame to the world frame
- We can find the coordinates of the point $\boldsymbol{a}_b = [\boldsymbol{a}_x^b, \boldsymbol{a}_y^b]^T$ from the body frame to the world frame, using the following equation: $\boldsymbol{a}_s = \boldsymbol{p} + \boldsymbol{R}\boldsymbol{a}_b$

Source: Modern Robotics: Mechanics, Planning, and Control, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press.

What is orientation (in 3D)?

The transformation that transforms a vector/point, $\boldsymbol{v}_b$, expressed in "body frame" to the same vector/point, but expressed in the "fixed/world frame" orientation, $\boldsymbol{v}_w$.

$$\boldsymbol{v}_w = \boldsymbol{R}_{wb}\boldsymbol{v}_b$$

where $\boldsymbol{R}_{\boldsymbol{wb}} \in \mathcal{SO}(3)$ is a **rotation matrix** that rotates the "fixed/world frame" such that it matches the "body frame". We use this rotation (transformation) matrix to rotate a vector from "body frame" to the "world frame".

Frames and Rotation Matrices

A frame is basically a set of 3 unit-norm vectors, that are orthogonal to each other:

$$\boldsymbol{w}_x \cdot \boldsymbol{w}_y = 0$$
$$\boldsymbol{w}_x \cdot \boldsymbol{w}_z = 0$$
$$\boldsymbol{w}_y \cdot \boldsymbol{w}_z = 0$$

where $\mathcal{W} = (\boldsymbol{w}_x, \boldsymbol{w}_y, \boldsymbol{w}_z)$ is the world frame. If we also define a body frame as $\mathcal{B}$, then we have:

$$\boldsymbol{R}_{wb} = \begin{bmatrix} (\boldsymbol{w}_x \cdot \boldsymbol{b}_x) & (\boldsymbol{w}_x \cdot \boldsymbol{b}_y) & (\boldsymbol{w}_x \cdot \boldsymbol{b}_z) \\ (\boldsymbol{w}_y \cdot \boldsymbol{b}_x) & (\boldsymbol{w}_y \cdot \boldsymbol{b}_y) & (\boldsymbol{w}_y \cdot \boldsymbol{b}_z) \\ (\boldsymbol{w}_z \cdot \boldsymbol{b}_x) & (\boldsymbol{w}_z \cdot \boldsymbol{b}_y) & (\boldsymbol{w}_z \cdot \boldsymbol{b}_z) \end{bmatrix}$$

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We refer to this version of derivation of the rotation matrix as the **Direction Cosine Method/Matrix** (DCM).

Rotation Matrices

- If $\boldsymbol{R}_1, \boldsymbol{R}_2$ are rotation matrices, then $\boldsymbol{R}_1\boldsymbol{R}_2$ is also a rotation matrix
- $(\boldsymbol{R}_1\boldsymbol{R}_2)\boldsymbol{R}_3 = \boldsymbol{R}_1(\boldsymbol{R}_2\boldsymbol{R}_3)$
- $\boldsymbol{R}\boldsymbol{I} = \boldsymbol{I}\boldsymbol{R} = \boldsymbol{R}$ ($\boldsymbol{I}$ is the identity matrix)
- $\boldsymbol{R}^{-1}\boldsymbol{R} = \boldsymbol{I}$
- $\boldsymbol{R}^{-1} = \boldsymbol{R}^T$, aka $\boldsymbol{R}_{wb}^T = \boldsymbol{R}_{bw}$
- $\det(\boldsymbol{R}) = 1$
- $\boldsymbol{R}_{az} = \boldsymbol{R}_{ab}\boldsymbol{R}_{bc}\boldsymbol{R}_{cd}\ldots\boldsymbol{R}_{yz}$

Rotations around Principal Axes

Rotation around x-axis:

$$\boldsymbol{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

Rotation around y-axis:

$$\boldsymbol{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

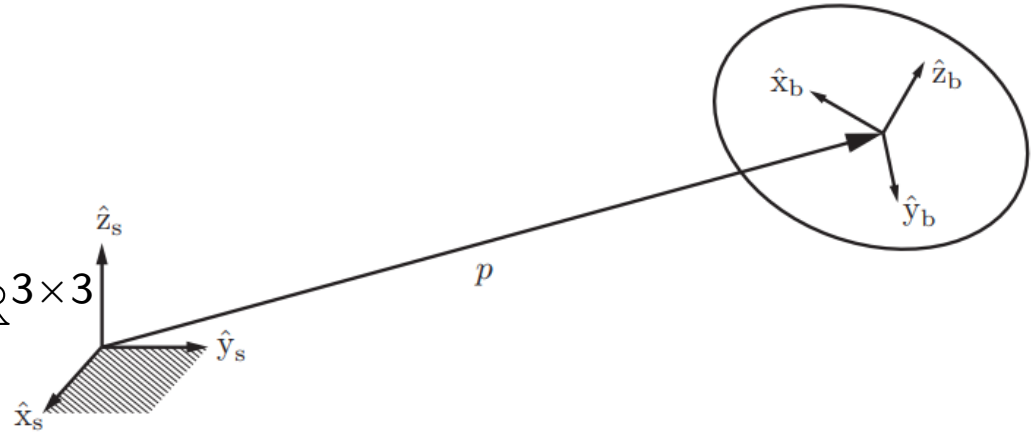
Rotation around z-axis:

$$\boldsymbol{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Full Pose Transformations

We have:

- $\boldsymbol{p}_s = \begin{bmatrix} p_x^s \\ p_y^s \\ p_z^s \end{bmatrix} \in \mathbb{R}^3$
- $\boldsymbol{R}_{sb} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \in \mathbb{R}^{3\times 3}$



- $(\boldsymbol{p}_s, \boldsymbol{R}_{sb})$ represent the transformation from the *b frame* to the *s frame*
- We can get the coordinates of a point $\boldsymbol{a}_b = [a_x^b, a_y^b, a_z^b]^T$ from the *b frame* to the *s frame*, using the following: $\boldsymbol{a}_s = \boldsymbol{p}_s + \boldsymbol{R}_{sb}\boldsymbol{a}_b$

Figure Source: Modern Robotics: Mechanics, Planning, and Control, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press.

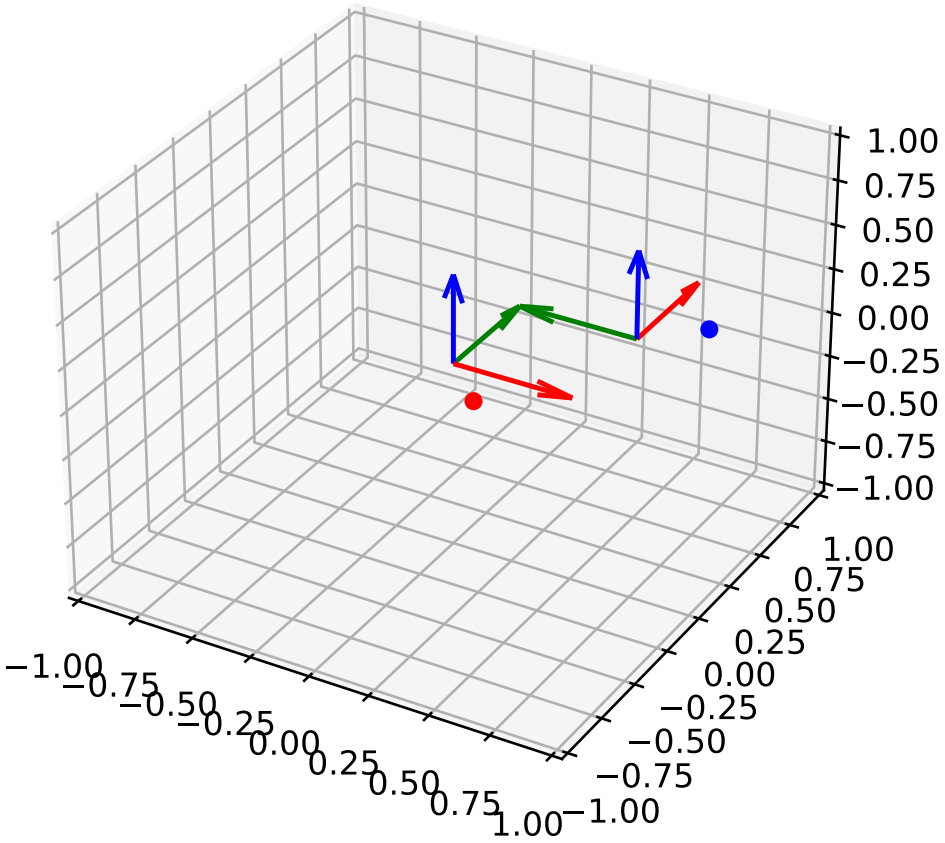
Homogeneous Transformations

- We assume that every point in 3D space can be defined as: $\boldsymbol{p} = [x, y, z, 1]$
- Whereas every vector as: $\boldsymbol{v} = [x, y, z, 0]$
- We define the Homogeneous Transformation Matrix $\boldsymbol{T} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ 0 & 1 \end{bmatrix}$, where $\boldsymbol{R}$ is a rotation matrix and $\boldsymbol{t}$ a translation vector
- T transforms point $\boldsymbol{p}$ as follows: $\boldsymbol{p}' = \boldsymbol{T}\boldsymbol{p} = \boldsymbol{R}\boldsymbol{p} + \boldsymbol{t}$
- $\boldsymbol{v}' = \boldsymbol{T}\boldsymbol{v} = \boldsymbol{R}\boldsymbol{v}$: vectors can only be rotated!

Homogeneous Transformation Matrices

- If $\boldsymbol{T}_1, \boldsymbol{T}_2$ are transformation matrices, then $\boldsymbol{T}_1\boldsymbol{T}_2$ is also a transformation matrix
- $(\boldsymbol{T}_1\boldsymbol{T}_2)\boldsymbol{T}_3 = \boldsymbol{T}_1(\boldsymbol{T}_2\boldsymbol{T}_3)$
- $\boldsymbol{T}\boldsymbol{I} = \boldsymbol{I}\boldsymbol{T} = \boldsymbol{T}$ (there exist the identity matrix)
- $\boldsymbol{T}^{-1}\boldsymbol{T} = \boldsymbol{I}$
- $\boldsymbol{T}^{-1} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{R}^T & -\boldsymbol{R}^T\boldsymbol{t} \\ 0 & 1 \end{bmatrix}$

Transformations Code Example



Bibliography

Chapter 3 from **Modern Robotics: Mechanics, Planning, and Control**, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press. ebook

Thank you

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